

An Order-Three Hénon Euler Superproduct: Exact Analytic Germ and First-Prime Conjugation Obstruction

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Abstract

We construct a genuinely non-scalar order-three symmetry reduction of the homogeneous area-preserving Hénon map. The full generating kernel $S_0(q, Q) = qQ + 2q^3$ is invariant under opposite cubic scalings of its two variables. Over every prime $p \equiv 1 \pmod{3}$, its Fourier-cubic quantization satisfies the reversing relation $U_p R_p = R_p^{-1} U_p$; hence the two-step time $T_p = U_p^2$ preserves three character sectors. The canonical integral augmentation weights $(2, -1, -1)$ yield a rational local superdeterminant with an exact chronological twisted-orbit trace formula. Smooth-leading-cubic estimates imply a uniform moment bound and a nonzero canonical Euler germ on $\operatorname{Re} s > 1$. Exact modular characteristic-polynomial calculations at all nine split primes through 73 show that virtual rank two does not produce bounded local spectral rank: the reduced numerator and denominator degrees reach 50 and 48 at $p = 73$. More decisively, an exact residue histogram at $p = 7$ proves that the first augmentation coefficient is nonreal. Thus the raw product fails conjugation symmetry even though every local factor is unitary and self-inversive. This supplies both a new Hénon Euler germ and a theorem-level obstruction to its promotion as a Riemann determinant. Companion rigidity controls close every finite Tate-CM repair. A viable successor must instead descend the conjugate-paired full Hénon-kernel moments to a self-dual compatible system over one fixed number field, with uniformly bounded rank and conductor.

1 Introduction

Finite-field quantizations of nonlinear maps offer an unusually direct meeting point between chronological dynamics, unitary spectra, and arithmetic Euler products. Their main danger is equally direct: a local unitary matrix can place zeros on a chosen vertical line without supplying the global symmetries of the completed Riemann zeta function. A useful candidate must therefore survive both construction and falsification gates.

This paper studies the homogeneous area-preserving Hénon map

$$H_0(q, p) = (-6q^2 - p, q),$$

generated by $S_0(q, Q) = qQ + 2q^3$. Direct cubic roots of the scalar chirp are geometrically constant and give no Hénon information. The full two-variable kernel, however, has a non-scalar order-three reversing symmetry. Its quantization exchanges the two nontrivial character sectors in one step and preserves all sectors in two steps. This forces an integral augmentation determinant rather than an arbitrary colored trace.

Our contributions are fourfold.

1. We prove the exact classical and quantum reversing relations and the resulting sector square identity.
2. We retain chronological Hénon time in a twisted path-sum formula and prove a locally uniform, zero-free canonical Euler germ on $\operatorname{Re} s > 1$.

3. We give exact modular certificates showing that the natural virtual rank-two factor retains linearly growing local spectral complexity.
4. We prove, without a zero table or floating-point root computation, that the first split-prime coefficient is nonreal. The raw Euler product is therefore rejected by the conjugation-symmetry gate of Route A.

The positive and negative conclusions are intentionally kept together. The order-three symmetry is genuine and reusable; the raw single-character Euler promotion is not. Companion controls also close finite Tate–CM repairs. The only serious successor is a source-native compatible system over one fixed coefficient field whose Frobenius powers recover the Hénon-kernel moments with uniformly bounded rank and conductor, not a post hoc symmetrization of this product.

2 Context and claim boundary

The initial Hénon program begins from an area-preserving model proposed as a dynamical framework for Riemann-zero exploration [4]. Earlier work in this repository quantized the inhomogeneous Hénon generating kernel by a finite Fourier transform times a cubic phase and used ordered kernel powers to recover Hénon critical equations. The general cohomological rank and weight bound for a polynomial with smooth leading form is classical, following Deligne’s exponential-sum theorem [2]. We reuse that theorem only as a uniform moment majorant; the new content is the Hénon-specific reversing symmetry, augmentation factor, and exact first-prime obstruction.

Finite symmetry groups can factor transfer-operator determinants into representation sectors. Such reductions are well developed for symmetric holomorphic iterated-function schemes and Selberg zeta functions [1]. Our setting differs in two ways. First, a single Hénon time step reverses rather than commutes with the cubic grading. Second, the arithmetic fibers have dimensions growing with p . An integral virtual representation gives a rational local determinant, but small virtual dimension does not imply small reduced numerator and denominator.

Twisted Artin–Schreier–Kummer sums admit broad cohomological estimates [3]. They also provide an essential control here. A Kummer lift of the homogeneous scalar phase is constant after the change of variables u/x , so any residual cubic arithmetic character is externally supplied and not generated by Hénon chronology. The present construction instead uses the symmetry of the full generating kernel.

A companion sequence of exact repository controls closes the simpler Eisenstein alternatives. A $j = 0$ CM elliptic curve supplies honest degree-two Frobenius factors, but they are auxiliary arithmetic geometry, not the Hénon orbit ledger. More decisively, within the finite virtual span of that CM H^1 and the two Tate pieces, equality with the Riemann local factor at three primes forces the CM coefficient to vanish. Thus the next gate cannot be another finite Tate–CM supercancellation; it must test descent of the source-native full-kernel moments themselves.

Finally, all claims are evaluated against a strict dynamical-determinant workflow. We prove an analytic Euler germ and local reciprocal duality, but we do not infer a completed functional equation, a Gamma factor, or a Hilbert–Pólya operator. The exact failure of same-point conjugation symmetry is treated as a stopping theorem for the raw candidate.

3 Reversing symmetry and the augmentation determinant

Let $p > 3$ satisfy $p \equiv 1 \pmod{3}$, and fix $\rho \in \mathbb{F}_p^\times$ of order three. On $\mathcal{H}_p = \ell^2(\mathbb{F}_p)$ define

$$(U_p f)(Q) = p^{-1/2} \sum_{q \in \mathbb{F}_p} \exp\left(\frac{2\pi i}{p}(qQ + 2q^3)\right) f(q), \quad (R_p f)(x) = f(\rho x).$$

The operator U_p is unitary because it is a normalized finite Fourier transform followed by a unitary chirp.

Theorem 3.1 (Intrinsic cubic reversing symmetry). *Let $g(q, p_0) = (\rho q, \rho^{-1} p_0)$. Then*

$$H_0 g = g^{-1} H_0, \quad U_p R_p = R_p^{-1} U_p.$$

Consequently $T_p = U_p^2$ commutes with R_p .

Proof. The classical identity follows by direct substitution. For the quantum identity, change the summation variable by $q = \rho^{-1} r$ and use $\rho^3 = 1$. Squaring the relation gives $[U_p^2, R_p] = 0$. $\square$

Let $\omega = e^{2\pi i/3}$ and $\mathcal{H}_{p,k} = \ker(R_p - \omega^k I)$. Multiplication by ρ has the fixed orbit $\{0\}$ and $(p-1)/3$ free orbits, giving

$$d_{p,0} = \frac{p+2}{3}, \quad d_{p,1} = d_{p,2} = \frac{p-1}{3}.$$

Moreover, U_p maps grade k to grade $-k$ and intertwines the restrictions of T_p . Thus $T_{p,1}$ and $T_{p,2}$ are unitarily equivalent.

Definition 3.2 (Integral augmentation factor). *For $D_{p,k}(z) = \det(I - zT_{p,k})$, set*

$$D_p^{\text{aug}}(z) = \frac{D_{p,0}(z)^2}{D_{p,1}(z)D_{p,2}(z)} = \left(\frac{D_{p,0}(z)}{D_{p,1}(z)} \right)^2.$$

The weights $(2, -1, -1)$ lie in the integral representation ring. This is an equivariant virtual determinant, not a Berezinian: a $\mathbb{Z}/3$ grading does not provide a canonical $\mathbb{Z}/2$ parity. Its virtual dimension is two, but the individual degrees are of order p .

Proposition 3.3 (Determinant normalization). *For $k = 0, 1, 2$,*

$$\det T_{p,k} = (-1)^{(p-1)/6}.$$

Consequently $D_{p,0}/D_{p,1} \sim -z$ and $D_p^{\text{aug}}(z) \sim z^2$ as $z \rightarrow \infty$.

The proof is given in appendix A.1. In particular, no cancellation can make the augmentation factor constant.

4 Chronological traces and an Euler half-plane

Let

$$P_k = \frac{1}{3} \sum_{a=0}^2 \omega^{-ka} R_p^a.$$

The representation-ring identity

$$2P_0 - P_1 - P_2 = R_p + R_p^2$$

gives the logarithmic moments

$$\begin{aligned} A_{p,n} &= 2 \operatorname{Tr}(T_{p,0}^n) - \operatorname{Tr}(T_{p,1}^n) - \operatorname{Tr}(T_{p,2}^n) \\ &= \operatorname{Tr}((R_p + R_p^2)U_p^{2n}) = 2 \operatorname{Tr}(R_p U_p^{2n}). \end{aligned} \tag{1}$$

The last equality follows from the unitary equivalence of the two twisted sectors.

Expanding the ordered matrix product, for $a = 1, 2$ and $N \geq 1$,

$$\mathrm{Tr}(R_p^a U_p^N) = p^{-N/2} \sum_{x_0, \dots, x_{N-1} \in \mathbb{F}_p} \exp\left(\frac{2\pi i}{p} \Phi_{N, \rho^a}(x)\right),$$

where

$$\Phi_{N, \rho^a} = \sum_{j=0}^{N-2} (x_j x_{j+1} + 2x_j^3) + \rho^a x_{N-1} x_0 + 2x_{N-1}^3.$$

This is a twisted closure $x_N = \rho^a x_0$. Its critical equations are the ordered Hénon recurrence, equivalently $H_0^N(z_0) = g^a z_0$. Chronology has not been replaced by an averaged transition matrix.

The highest homogeneous part of $\Phi_{2n, \rho}$ is $2 \sum_j x_j^3$. Its gradient $(6x_j^2)_j$ has no common projective zero for $p > 3$. Deligne's smooth-leading-form estimate therefore yields

$$|\mathrm{Tr}(R_p U_p^{2n})| \leq 4^n, \quad |A_{p,n}| \leq 2 \cdot 4^n.$$

Theorem 4.1 (Hénon augmentation Euler germ). *The product*

$$\mathcal{D}_{\mu_3}(s) = \prod_{\substack{p > 3 \\ p \equiv 1 \pmod{3}}} D_p^{\mathrm{aug}}(p^{-s})$$

has a canonical, locally uniformly convergent trace logarithm and is holomorphic and nonzero on $\mathrm{Re} s > 1$.

Proof. For $\sigma = \mathrm{Re} s > 1$,

$$\sum_p \sum_{n \geq 1} \frac{|A_{p,n}|}{n} p^{-n\sigma} \leq 2 \sum_p \sum_{n \geq 1} \frac{(4p^{-\sigma})^n}{n} < \infty.$$

The bound is locally uniform on closed sub-half-planes. □

Since every $T_{p,k}$ is unitary, the reduced local zeros and poles lie on $|z| = 1$. For the canonical substitution $z = p^{-s}$ this is $\mathrm{Re} s = 0$. In fact, with $G_p = D_{p,0}/D_{p,1}$,

$$G_p(z) = -z \overline{G_p(1/\bar{z})}, \quad D_p^{\mathrm{aug}}(z) = z^2 \overline{D_p^{\mathrm{aug}}(1/\bar{z})}.$$

The optional display coordinate $z = p^{1/2-s}$ places the local divisors on $\mathrm{Re} s = 1/2$ but shifts the proved convergence domain to $\mathrm{Re} s > 3/2$. This local self-inversivity is not a global functional equation: the formal product of the resulting factors p^{1-2s} diverges, and no conductor or Gamma completion has been constructed.

5 Exact noncancellation certificate

The assertion “virtual rank two” could conceal a large common spectral factor. We test this without floating-point eigenvalues. For every frozen prime p , choose the smallest auxiliary prime $\ell \equiv 1 \pmod{3p}$. Then $\mathbb{F}_\ell$ contains exact roots of unity of orders p and three. Because $T_p = U_p^2$, the normalization is p^{-1} and no square root is required.

We construct the three sector bases, restrict T_p , and compute exact characteristic polynomials over $\mathbb{F}_\ell$. A coprime reduction at a good prime proves that the characteristic-zero polynomials were coprime. The producer uses FLINT; the checker independently uses integer matrix arithmetic, Faddeev–LeVerrier characteristic polynomials, and Euclidean polynomial gcds.

Each row is an exact theorem about that characteristic-zero local factor. The table is not promoted to an all-prime coprimality theorem. It establishes the narrower and decisive point:

Table 1: Frozen exact coprimality controls. The two nontrivial sector polynomials are equal; the trivial and nontrivial polynomials are coprime.

p	ℓ	d_0	d_1	reduced degrees (N, D)
7	43	3	2	(6, 4)
13	79	5	4	(10, 8)
19	229	7	6	(14, 12)
31	373	11	10	(22, 20)
37	223	13	12	(26, 24)
43	1033	15	14	(30, 28)
61	367	21	20	(42, 40)
67	1609	23	22	(46, 44)
73	439	25	24	(50, 48)

order-three symmetry alone does not enforce a bounded-rank cancellation mechanism. Any proposed global continuation must explain increasingly large local divisors rather than appeal to the net degree two.

As a scalar control, the cover $u^3 = 2x^3$ becomes $t^3 = 2$ after $t = u/x$. Its geometric scaling local system is constant; a residual constant-field Frobenius factor carries no x -dependent Hénon chronology. Thus the full kernel symmetry used above is essential.

6 A first-prime conjugation obstruction

Local unitary divisors and reciprocal identities are not enough for a Riemann determinant. The raw product must also be of real type. This fails at the first split prime.

Theorem 6.1 (Exact $p = 7$ obstruction). *For the raw single-additive-character augmentation product, $A_{7,1}$ is nonreal. Consequently*

$$\mathcal{D}_{\mu_3}(\bar{s}) \not\equiv \overline{\mathcal{D}_{\mu_3}(s)}.$$

Thus global real-type symmetry is refuted on the defining half-plane.

Proof. Take $\rho = 2 \in \mathbb{F}_7$ and $\xi = e^{2\pi i/7}$. Direct exact enumeration of the twisted phase

$$2x^3 + 2y^3 + 3xy$$

gives the residue histogram

$$(4, 9, 18, 3, 6, 6, 3).$$

Thus

$$A_{7,1} = \frac{2}{7}(4 + 9\xi + 18\xi^2 + 3\xi^3 + 6\xi^4 + 6\xi^5 + 3\xi^6).$$

The coefficient vector of $A_{7,1} - \overline{A_{7,1}}$ is

$$(0, 6, 12, -3, 3, -12, -6).$$

If its degree-at-most-six polynomial vanished at ξ , it would be a scalar multiple of the minimal polynomial $\Phi_7(X) = 1 + X + \cdots + X^6$. The coefficients are not constant, so the difference is nonzero.

Finally, $D_7^{\text{aug}}(z) = 1 - A_{7,1}z + O(z^2)$. In the absolutely convergent global Dirichlet expansion, the coefficient carrying the unique prime label seven cannot be canceled by another prime. Hence same-point conjugation symmetry fails globally. $\square$

Replacing ρ by ρ^{-1} does not help: the two twisted traces are equal by theorem 3.1. Multiplying by the conjugate local factor would repair real type, but it doubles the already growing local degrees and gives neither a global functional equation nor a canonical completion. It is a new candidate, not a correction to the object tested here.

7 Route-A decision and next gate

The construction makes a substantial positive step: a non-scalar symmetry of the Hénon generating kernel forces a chronological sector determinant, and that determinant has a proved Euler half-plane. It also meets a clean stopping theorem before any comparison with Riemann zeros.

The formal Route-A tuple is

(A1_WEAK, A2_ANALYTIC_DETERMINANT, A3_FAIL, A4_NATURAL_QUANTIZATION),
with overall status `ROUTE_A_REJECTED`. A1 is weak because primes index different arithmetic fibers rather than primitive orbits of one real map. A2 records the explicit rational local determinants and the proved nonzero half-plane; it is not a divisor match. A3 fails by theorem 6.1. A4 is natural at every finite place, but no single global Hilbert–Pólya operator has been constructed. Route B is not authorized.

The companion C41–C42 rigidity chain already closes the obvious finite-dimensional repair: adjoining the $j = 0$ CM elliptic motive and Tate pieces matches the Riemann local factor only when the CM contribution is deleted. The next large gate is therefore precise. After pairing the additive character with its inverse and including split and inert places, do the full-kernel moment sequences descend to a pure self-dual compatible system over one fixed coefficient field, with rank and conductor bounded independently of p ? A fixed coefficient field first bounds the algebraic degrees of the paired first moments. Only if that descent gate survives does fixed rank force a uniform recurrence, equivalently bounded Hankel rank, for all chronological moments. The frozen coprimality ledger points toward growth, but does not settle either paired all-prime statement. A positive answer must construct the compatible system and functional equation before inspecting zeros; unbounded trace degree or Hankel rank closes the order-three Hénon Euler route. Fitted conjugate factors are not admissible.

References

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A Supplementary proofs and reproducibility

A.1 Determinant normalization

Write $U = FM$, where F is normalized Fourier transform and $M(q) = e_p(2q^3)$. On every nonzero μ_3 orbit the chirp is scalar. The product of these scalars is one, because the orbits are indexed

by the cube subgroup and the sum of that subgroup vanishes. The extra vector at zero in grade zero also has eigenvalue one. Therefore $\det(M|\mathcal{H}_{p,k}) = 1$ in every grade.

Furthermore $F^2 = J$, where $(Jf)(x) = f(-x)$. Since $p \equiv 1 \pmod{6}$, -1 is a cube. Put $m = (p-1)/3$; negation pairs the m nonzero cubic orbits into $m/2$ transpositions. The zero vector is fixed in grade zero. Hence

$$\det(J|\mathcal{H}_{p,k}) = (-1)^{m/2} = (-1)^{(p-1)/6}$$

for every k . Block determinants for U^2 give proposition 3.3.

Complex conjugation gives

$$\bar{U} = F^{-1}M^{-1} = MU^{-1}M^{-1}.$$

Because M commutes with R , conjugation sends grade k to grade $-k$ and gives cross-grade similarity between $\bar{T}_k$ and T_{-k}^{-1} . Grade zero is self-dual. For grades one and two, combine this statement with the independently proved equality $D_{p,1} = D_{p,2}$. This proves the local self-inversive relations used in the text without identifying conjugate grades silently.

A.2 Exact arithmetic protocol

The frozen producer and checker share only scalar conventions and released JSON. The producer uses Python–FLINT. The checker does not import the producer or FLINT; it independently reconstructs matrices, computes characteristic polynomials by the Faddeev–LeVerrier recurrence, and computes polynomial gcds by Euclid’s algorithm. All twelve semantic blocks are compared recursively with type-strict JSON equality. The release runner reproduces the certificate in a temporary directory before checking the frozen copy.